

# 1. Table Definitions

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### 1.1. all\_files

Master list of filenames seen in the collect. Derived from process executions, dll loads and file usage activity.

| Column           | Data Type | Description                                                                                            |
|------------------|-----------|--------------------------------------------------------------------------------------------------------|
| dll_num_rows     | DOUBLE    | Number of times used as a DLL                                                                          |
| file_num_rows    | DOUBLE    | Number of times referenced in file read/write activity                                                 |
| filename         | STRING    | Fully qualified filename                                                                               |
| num_hosts        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| process_num_rows | DOUBLE    | Number of rows file was executed as a process                                                          |

### 1.2. files

Files by hostname with summary information for executions, dlls and file usage.

| Column              | Data Type | Description                                                     |
|---------------------|-----------|-----------------------------------------------------------------|
| dll_first_seen      | TIMESTAMP | Earliest time seen                                              |
| dll_last_seen       | TIMESTAMP | Latest time seen                                                |
| dll_num_rows        | DOUBLE    | Number of times used as a DLL                                   |
| file_first_seen     | TIMESTAMP | Earliest time seen                                              |
| file_id             | STRING    | Unique ID for a file. Hash of hostname+fully qualified filename |
| file_last_seen      | TIMESTAMP | Latest time seen                                                |
| file_num_rows       | DOUBLE    | Number of times referenced in file read/write activity          |
| filename            | STRING    | Fully qualified filename                                        |
| hostname            | STRING    | Hostname data collected from                                    |
| max_process_term    | TIMESTAMP | Maximum value in the interval                                   |
| min_process_started | TIMESTAMP | Minimum value in the interval                                   |
| process_num_rows    | DOUBLE    | Number of rows file was executed as a process                   |

### 1.3. host

Unique entry for every instrumented host.

| Column          | Data Type | Description                                                                                            |
|-----------------|-----------|--------------------------------------------------------------------------------------------------------|
| Hostname        | STRING    | Hostname data collected from                                                                           |
| ad_domain       | STRING    | Active directory host is joined to                                                                     |
| agent_ids       | STRING[]  | List of agent ids. A long running host may have several dur to update, reinstalls, etc.                |
| arch            | STRING    | Chip architecture                                                                                      |
| domain_role     | STRING    | Host role within the domain. Ex: domain controller, member, etc.                                       |
| etl_version     | STRING    | Wintap data collection plugin version                                                                  |
| first_seen      | TIMESTAMP | Earliest time seen                                                                                     |
| has_battery     | BOOLEAN   | Does this host have a battery? Indicates laptop if true                                                |
| last_boot       | INT64     | Time of last boot, in epoch seconds                                                                    |
| last_seen       | TIMESTAMP | Latest time seen                                                                                       |
| num_ad_domain   | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_arch        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_domain_role | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |

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TABLE 3 – Continued from previous page

| Column              | Data Type | Description                                                                                            |
|---------------------|-----------|--------------------------------------------------------------------------------------------------------|
| num_etl_version     | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_has_battery     | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_last_boot       | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_os              | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_os_version      | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_processor_count | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_processor_speed | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_rows            | INT64     | Number of row behind this summary                                                                      |
| num_wintap_version  | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| os                  | STRING    | Host operating system                                                                                  |
| os_family           | STRING    | OS Family: windows, linux, osx                                                                         |
| os_version          | STRING    | Operating System Version                                                                               |
| processor_count     | INTEGER   | Number of processors cores                                                                             |
| processor_speed     | INT64     | Speed of processors                                                                                    |
| wintap_version      | STRING    | Version Wintap core. Each plugin has its own version.                                                  |

#### 1.4. host\_ip

Network interface information for hosts.

| Column          | Data Type | Description                                                                                             |
|-----------------|-----------|---------------------------------------------------------------------------------------------------------|
| Hostname        | STRING    | Hostname data collected from                                                                            |
| MTU             | INTEGER   | Should be max transmtion unit, but its currently not collected.                                         |
| agent_id        | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| first_seen      | TIMESTAMP | Earliest time seen                                                                                      |
| interface       | STRING    | Should be the interface device name, but its currently not collected.                                   |
| ip_addr         | INT64     | An IP address on the host. Stored as an integer                                                         |
| ip_addr_no      | STRING    | An IP address on the host. In dotted quad notation.                                                     |
| last_seen       | TIMESTAMP | Latest time seen                                                                                        |
| mac             | STRING    | Mac address as string (colon delimited)                                                                 |
| num_rows        | INT64     | Number of row behind this summary                                                                       |
| os_family       | STRING    | OS Family: windows, linux, osx                                                                          |
| private_gateway | STRING    | IP of the gateway used by this interface                                                                |

#### 1.5. labels\_graph\_net\_conn

Summarizes network-related labels by conn\_id

| Column                     | Data Type | Description                                                                                                                                                                                                                                                              |
|----------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| conn_id                    | STRING    | Hash of "normalized" 5 tuple (ip1, port1, ip2, port2, L4 protocol). To represent A to B the same as B to A, the lower IP (as int) along with its port comes first in the hash preimage ordering. When ip1 == ip2, then the pair order is decided by lowest port instead. |
| label_num_hits             | INT64     | Number of unique labels hit                                                                                                                                                                                                                                              |
| label_num_sources          | INT64     | Number of unique sources of labels                                                                                                                                                                                                                                       |
| label_num_uniq_annotations | INT64     | Number of unique annotations                                                                                                                                                                                                                                             |
| label_source               | STRING    | For now, only a single source: "networkx"                                                                                                                                                                                                                                |

## 1.6. labels\_graph\_nodes

Parse JSON in NetworkX files into a flat structure representing the nodes

| Column     | Data Type | Description                                                                                                                                     |
|------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| annotation | STRING    | Annotations are manually added to specific nodes by the person creating the networkx graph of activity. Used to further highlight the activity. |
| filename   | STRING    | Fully qualified filename                                                                                                                        |
| id         | STRING    | Unique ID for the node in the graph. Often a PID_HASH, CONN_ID or similar.                                                                      |
| label      | STRING    | Graph node label. Often a PROCESS_NAME or similar.                                                                                              |
| node_type  | STRING    | Type of node: PROCESS, NETWORK, FILE, etc.                                                                                                      |

## 1.7. labels\_graph\_process\_summary

Summarizes process-related labels by pid\_hash

| Column                     | Data Type | Description                                                               |
|----------------------------|-----------|---------------------------------------------------------------------------|
| label_num_hits             | INT64     | Number of unique labels hit                                               |
| label_num_sources          | INT64     | Number of unique sources of labels                                        |
| label_num_uniq_annotations | INT64     | Number of unique annotations                                              |
| label_source               | STRING    | For now, only a single source: "networkx"                                 |
| pid_hash                   | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time) |

## 1.8. labels\_networkx

Import of networkx graph data. Used primarily as base to build summary tables. Data in this table is JSON, tricky to work with directly.

| Column        | Data Type | Description              |
|---------------|-----------|--------------------------|
| directed      | BOOLEAN   | Directed graph or not.   |
| filename      | STRING    | Fully qualified filename |
| is_multigraph | BOOLEAN   | Multigraph or not        |
| links         | STRUCT    | List of links            |
| nodes         | STRUCT    | List of nodes            |

## 1.9. lolbas

Joins process data with LOLBAS (Living Off the Land Binaries and Scripts) data. Summarizes LOLBAS-related activity by pid\_hash.

| Column                | Data Type | Description                                                                 |
|-----------------------|-----------|-----------------------------------------------------------------------------|
| acknowledgements      | STRING    | Person who submitted the entry. May be more than 1.                         |
| author                | STRING    | Person or group who submitted the entry.                                    |
| command               | STRING    | The command with arguments                                                  |
| command_category      | STRING    | Intent. Ex: EXECUTE, UAC BYPASS, CONCEAL, etc.                              |
| command_description   | STRING    | Description of the command                                                  |
| command_privileges    | STRING    | Required privileges to execute                                              |
| command_usecase       | STRING    | Description of use case for attacker                                        |
| date                  | DATE      | Date created                                                                |
| description           | STRING    | Normal purpose of the command                                               |
| detections            | STRING    | How to detect exploit: Many are URLs pointing to SIGMA rule, Defender, etc. |
| filename              | STRING    | Filename of executable. No path.                                            |
| mitre_attck_technique | STRING    | Specific MITRE attck code. Ex: T1003                                        |
| operating_system      | STRING    | OSes binary is found on. Some very specific, some general.                  |
| paths                 | STRING    | Paths where binary is found. More than one are comma separated.             |
| resources             | STRING    | URL to more info                                                            |
| url                   | STRING    | URL to LolBAS project about the exploit                                     |

## 1.10. mitre\_labels

| Column      | Data Type | Description                                                                                                                                                 |
|-------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| analytic_id | STRING    | Mitre analytic ID. Ex: CAR-2021-05-012                                                                                                                      |
| entity      | STRING    | Wintap entity ID (hash). The type of entity (PID_HASH, CONN_ID, etc) is defined in the ENTITY_TYPE column. Note: Currently, only PID_HASHes are identified. |
| entity_type | STRING    | Wintap entity type. Determines what kind of hash is in the ENTITY column. Ex: PID_HASH, CONN_ID, FILE_ID                                                    |
| time        | TIMESTAMP | Timestamp of activity                                                                                                                                       |

## 1.11. process

Each row is a process execution. Processes are the central object of this entire data model.

| Column          | Data Type | Description                                                                                                                   |
|-----------------|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| agent_id        | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install.                       |
| args            | STRING    | Command line arguments                                                                                                        |
| commit_charge   | INT64     | Commit charge is the total amount of virtual memory guaranteed for all processes to fit in physical memory and the page file. |
| commit_peak     | INT64     | Total memory committed                                                                                                        |
| cpu_cycle_count | INT64     | Always zero (note: more than zero cycles were generally used in a process)                                                    |

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TABLE 11 – Continued from previous page

| Column                  | Data Type | Description                                                                                                                                                                                                                                                                                                |
|-------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cpu_utilization         | INTEGER   | Ignore: unable to collect. This *should* be a value indicating the percentage of cpu utilized by the process over its life, but turns out be harder to collect than expected.                                                                                                                              |
| exit_code               | INT64     | Process exit code                                                                                                                                                                                                                                                                                          |
| file_id                 | STRING    | Unique ID for a file. Hash of hostname+fully qualified filename                                                                                                                                                                                                                                            |
| file_md5                | STRING    | MD5 hash of the binary process/dll                                                                                                                                                                                                                                                                         |
| file_sha2               | STRING    | SHA2 hash of the file contents                                                                                                                                                                                                                                                                             |
| filename                | STRING    | Fully qualified filename                                                                                                                                                                                                                                                                                   |
| first_seen              | TIMESTAMP | Earliest time seen                                                                                                                                                                                                                                                                                         |
| hard_fault_count        | INTEGER   | Count of hard page faults                                                                                                                                                                                                                                                                                  |
| hostname                | STRING    | Hostname data collected from                                                                                                                                                                                                                                                                               |
| last_seen               | TIMESTAMP | Latest time seen                                                                                                                                                                                                                                                                                           |
| num_agent_id            | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_args                | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_file_md5            | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_file_sha2           | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_parent_os_pid       | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_parent_pid_hash     | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_process_name        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_process_path        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_process_start       | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_process_stop        | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| num_user_name           | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                                                     |
| os_family               | STRING    | OS Family: windows, linux, osx                                                                                                                                                                                                                                                                             |
| os_pid                  | INTEGER   | Operating system process id                                                                                                                                                                                                                                                                                |
| parent_os_pid           | INTEGER   | OS Process ID of parent process                                                                                                                                                                                                                                                                            |
| parent_pid_hash         | STRING    | PID_HASH of parent process                                                                                                                                                                                                                                                                                 |
| pid_hash                | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                                                                                                                                                                                                                                  |
| process_name            | STRING    | Filename of executable                                                                                                                                                                                                                                                                                     |
| process_path            | STRING    | Fully qualified path of the process executable                                                                                                                                                                                                                                                             |
| process_started         | TIMESTAMP | Process start timestamp                                                                                                                                                                                                                                                                                    |
| process_started_seconds | DOUBLE    | Process start time in unix epoch seconds                                                                                                                                                                                                                                                                   |
| process_stop_seconds    | DOUBLE    | Process stop in unix epoch seconds                                                                                                                                                                                                                                                                         |
| process_term            | TIMESTAMP | Process termination timestamp                                                                                                                                                                                                                                                                              |
| read_operation_count    | INT64     | Count of number of read operations                                                                                                                                                                                                                                                                         |
| read_transfer_kilobytes | INT64     | Number of K bytes read                                                                                                                                                                                                                                                                                     |
| token_elevation_type    | INTEGER   | Indicates UAC privilege status: 1 (Default) = no linked token, UAC disabled or non-admin account; 2 (Full) = elevated administrator with full rights after UAC approval; 3 (Limited) = filtered admin token before elevation with available full token; 4 = undocumented value, treat as unknown/reserved. |
| user_name               | STRING    | Username. Note that on windows there are many usernames that represent system or other background activity                                                                                                                                                                                                 |

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TABLE 11 – Continued from previous page

| Column                   | Data Type | Description                         |
|--------------------------|-----------|-------------------------------------|
| write_operation_count    | INT64     | Count of number of write operations |
| write_transfer_kilobytes | INT64     | Number of K bytes written           |

## 1.12. process\_conn\_incr

Network connection increments for each process. (KEY: PID\_HASH + CONN\_ID + INCR\_START\_SECS)

| Column               | Data Type | Description                                                                                                                                                                                                                                                              |
|----------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hostname             | STRING    | Hostname data collected from                                                                                                                                                                                                                                             |
| agent_id             | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install.                                                                                                                                                                  |
| conn_id              | STRING    | Hash of "normalized" 5 tuple (ip1, port1, ip2, port2, L4 protocol). To represent A to B the same as B to A, the lower IP (as int) along with its port comes first in the hash preimage ordering. When ip1 == ip2, then the pair order is decided by lowest port instead. |
| first_seen           | TIMESTAMP | Earliest time seen                                                                                                                                                                                                                                                       |
| incr_start           | TIMESTAMP | Start time of a 1 minute increment used to aggregate the high-volume events                                                                                                                                                                                              |
| last_seen            | TIMESTAMP | Latest time seen                                                                                                                                                                                                                                                         |
| local_ip_addr        | STRING    | IP address on the host collecting data. This address is local to the host sensor. In dotted quad notation.                                                                                                                                                               |
| local_ip_int         | INT64     | IP of the host collecting this data as 32-bit int                                                                                                                                                                                                                        |
| local_port           | INTEGER   | Port of the process on the collecting host                                                                                                                                                                                                                               |
| max_10sec_eventcount | INTEGER   | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_size             | INT64     | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_tcp_recv_count   | INTEGER   | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_tcp_recv_size    | INT64     | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_tcp_send_count   | INTEGER   | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_tcp_send_size    | INT64     | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_udp_recv_count   | INTEGER   | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_udp_recv_size    | INT64     | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_udp_send_count   | INTEGER   | Maximum value in the interval                                                                                                                                                                                                                                            |
| max_udp_send_size    | INT64     | Maximum value in the interval                                                                                                                                                                                                                                            |
| min_10sec_eventcount | INTEGER   | Minimum value in the interval                                                                                                                                                                                                                                            |
| min_size             | INT64     | Minimum value in the interval                                                                                                                                                                                                                                            |
| min_tcp_recv_size    | INT64     | Minimum value in the interval                                                                                                                                                                                                                                            |
| min_tcp_send_size    | INT64     | Minimum value in the interval                                                                                                                                                                                                                                            |
| min_udp_recv_size    | INT64     | Minimum value in the interval                                                                                                                                                                                                                                            |
| min_udp_send_size    | INT64     | Minimum value in the interval                                                                                                                                                                                                                                            |
| num_raw_rows         | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                   |
| os_family            | STRING    | OS Family: windows, linux, osx                                                                                                                                                                                                                                           |
| pid_hash             | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                                                                                                                                                                                                |
| process_name         | STRING    | Filename of executable                                                                                                                                                                                                                                                   |
| protocol             | STRING    | Layer 4 protocol (TCP, UDP)                                                                                                                                                                                                                                              |
| remote_ip_addr       | STRING    | IP address the host collecting data is talking to. This address is remote to the host sensor. In dotted quad notation.                                                                                                                                                   |
| remote_ip_int        | INT64     | IP of the remote host for this connection increment as 32-bit int                                                                                                                                                                                                        |
| remote_port          | INTEGER   | Port of the remote host for this connection increment                                                                                                                                                                                                                    |
| sq_size              | DOUBLE    | Squared size of bytes in network packet. Ask Chris B!                                                                                                                                                                                                                    |

Continued on next page

TABLE 12 – Continued from previous page

| Column               | Data Type | Description                                                              |
|----------------------|-----------|--------------------------------------------------------------------------|
| sq_tcp_rcv_size      | DOUBLE    | Square of the values in the interval. (Ask Chris B)                      |
| sq_tcp_send_size     | DOUBLE    | Square of the values in the interval. (Ask Chris B)                      |
| sq_udp_rcv_size      | DOUBLE    | Square of the values in the interval. (Ask Chris B)                      |
| sq_udp_send_size     | DOUBLE    | Square of the values in the interval. (Ask Chris B)                      |
| tcp_accept_count     | DOUBLE    | Number of TCP ACCEPT events on this connection for the time window       |
| tcp_connect_count    | DOUBLE    | Number of TCP CONNECT events on this connection for the time window      |
| tcp_disconnect_count | DOUBLE    | Number of TCP DISCONNECT events on this connection for the time window   |
| tcp_reconnect_count  | DOUBLE    | Number of TCP RECONNECT events on this connection for the time window    |
| tcp_rcv_count        | DOUBLE    | Number of TCP RECV events on this connection for the time window         |
| tcp_rcv_size         | DOUBLE    | Number of TCP RECV bytes received on this connection for the time window |
| tcp_retransmit_count | DOUBLE    | Number of TCP RETRANSMIT events on this connection for the time window   |
| tcp_send_count       | DOUBLE    | Number of TCP SEND events on this connection for the time window         |
| tcp_send_size        | DOUBLE    | Number of TCP bytes sent for the time window                             |
| tcp_tcpcopy_count    | DOUBLE    | Number of TCP TCPCOPY events on this connection for the time window      |
| tcp_tcpcopy_size     | DOUBLE    | Number of TCP TCPCOPY bytes on this connection for the time window       |
| total_events         | DOUBLE    | Sum of events counts for this connection increment                       |
| total_size           | DOUBLE    | Number of bytes observed for this connection increment                   |
| udp_rcv_count        | DOUBLE    | Number of UDP RECV events on this connection for the time window         |
| udp_rcv_size         | DOUBLE    | Number of UDP bytes received for the time window                         |
| udp_send_count       | DOUBLE    | Number of UDP SEND events on this connection for the time window         |
| udp_send_size        | DOUBLE    | Number of UDP bytes sent for the time window                             |

### 1.13. process\_exe\_file\_summary

Summary of files used in process executions. Derived from all process executions resulting in a single row per file per host.

| Column              | Data Type | Description                                                     |
|---------------------|-----------|-----------------------------------------------------------------|
| file_id             | STRING    | Unique ID for a file. Hash of hostname+fully qualified filename |
| filename            | STRING    | Fully qualified filename                                        |
| hostname            | STRING    | Hostname data collected from                                    |
| max_process_term    | TIMESTAMP | Maximum value in the interval                                   |
| min_process_started | TIMESTAMP | Minimum value in the interval                                   |
| process_num_rows    | INT64     | Number of rows file was executed as a process                   |
| source              | STRING    | Source of file: currently only PROCESS                          |

### 1.14. process\_file

File activity of processes. Activity is summarized to the PROCESS + FILE level. See RAW\_PROCESS\_FILE for activity detailed activity.

| Column          | Data Type | Description                                                                                             |
|-----------------|-----------|---------------------------------------------------------------------------------------------------------|
| Hostname        | STRING    | Hostname data collected from                                                                            |
| activity_type   | STRING    | File activity type. Ex: READ, WRITE                                                                     |
| agent_id        | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| bytes_requested | DOUBLE    | Total bytes requested                                                                                   |
| event_count     | DOUBLE    | Total ETW events                                                                                        |
| file_hash       | STRING    | MD5 hash of the file                                                                                    |
| file_id         | STRING    | Unique ID for a file. Hash of hostname+fully qualified filename                                         |
| filename        | STRING    | Name of the file                                                                                        |
| first_seen      | TIMESTAMP | Earliest time seen                                                                                      |
| last_seen       | TIMESTAMP | Latest time seen                                                                                        |
| max_event       | TIMESTAMP | Maximum value in the interval                                                                           |
| min_event       | TIMESTAMP | Minimum value in the interval                                                                           |
| num_raw_rows    | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| pid_hash        | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                               |
| process_name    | STRING    | Filename of executable                                                                                  |

### 1.15. process\_file\_summary

File activity summarize to the process level.

| Column             | Data Type | Description                                                                                             |
|--------------------|-----------|---------------------------------------------------------------------------------------------------------|
| Close_Events       | DOUBLE    | Number of file close events                                                                             |
| Create_Events      | DOUBLE    | Number of file create events                                                                            |
| Delete_Events      | DOUBLE    | Number of file delete events                                                                            |
| Hostname           | STRING    | Hostname data collected from                                                                            |
| Read_Bytes         | DOUBLE    | File bytes read                                                                                         |
| Read_Events        | DOUBLE    | Number of file read events                                                                              |
| Rename_Events      | DOUBLE    | Number of file rename events                                                                            |
| SetInfo_Events     | DOUBLE    |                                                                                                         |
| Write_Bytes        | DOUBLE    | File bytes written                                                                                      |
| Write_Events       | DOUBLE    | Number of file write events                                                                             |
| agent_id           | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| first_seen         | TIMESTAMP | Earliest time seen                                                                                      |
| last_seen          | TIMESTAMP | Latest time seen                                                                                        |
| num_null_filename  | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| num_raw_rows       | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| num_uniq_file_hash | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| pid_hash           | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                               |
| process_name       | STRING    | Filename of executable                                                                                  |

### 1.16. process\_image\_load

DLLs loaded and unloaded by process.

| Column         | Data Type | Description                                                                                             |
|----------------|-----------|---------------------------------------------------------------------------------------------------------|
| agent_id       | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| build_time     | INT64     | Build time of DLL                                                                                       |
| checksum       | INTEGER   | A checksum on the file contents of the DLL                                                              |
| default_base   | STRING    | Default base memory address                                                                             |
| file_id        | STRING    | Unique ID for a file. Hash of hostname+fully qualified filename                                         |
| file_md5       | STRING    | MD5 hash of the binary process/dll                                                                      |
| filename       | STRING    | Filename of the loaded code                                                                             |
| first_seen     | TIMESTAMP | Earliest time seen                                                                                      |
| hostname       | STRING    | Hostname data collected from                                                                            |
| image_base     | STRING    | Actual base memory address(?)                                                                           |
| last_seen      | TIMESTAMP | Latest time seen                                                                                        |
| max_image_size | INTEGER   | Maximum value in the interval                                                                           |
| min_image_size | INTEGER   | Minimum value in the interval                                                                           |
| num_load       | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| num_unload     | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| pid_hash       | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                               |
| process_name   | STRING    | Filename of executable                                                                                  |

### 1.17. process\_image\_load\_summary

Summarizes DLL and image load activity by process (pid\_hash).

| Column         | Data Type | Description                                                                                             |
|----------------|-----------|---------------------------------------------------------------------------------------------------------|
| agent_id       | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| dlls           | STRING[]  | List of DDL names as an array                                                                           |
| first_seen     | TIMESTAMP | Earliest time seen                                                                                      |
| hostname       | STRING    | Hostname data collected from                                                                            |
| last_seen      | TIMESTAMP | Latest time seen                                                                                        |
| num_uniq_files | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| pid_hash       | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                               |
| process_name   | STRING    | Filename of executable                                                                                  |

### 1.18. process\_lolbas\_summary

| Column          | Data Type | Description                                                               |
|-----------------|-----------|---------------------------------------------------------------------------|
| lolbas_cats     | STRING[]  | List categories from hits                                                 |
| lolbas_mitre    | STRING[]  | List of Mitre codes from hits                                             |
| lolbas_num_rows | INT64     | Number of hits                                                            |
| lolbas_privs    | STRING[]  | List of privileges from hist                                              |
| pid_hash        | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time) |

### 1.19. process\_mitre\_summary

| Column                    | Data Type  | Description                                                                |
|---------------------------|------------|----------------------------------------------------------------------------|
| mitre_analytic_ids        | STRING[]   | List of analytic IDs from hits                                             |
| mitre_analytic_types      | STRING[][] | List of analytic types from hits                                           |
| mitre_information_domains | STRING[]   | List domains from hits. Ex: Analytic, Host, Network                        |
| mitre_num_rows            | INT64      | Number of hits                                                             |
| mitre_subtypes            | STRING[][] | List of subtypes from hits. Ex: Map building, Anomaly, Host-flow, Process. |
| pid_hash                  | STRING     | Globally unique process hash (hash from hostname, os pid, and start_time)  |

### 1.20. process\_net\_conn

Network connections for each process. These are summarize to the PROCESS and Network 5-tuple (KEY: PID\_HASH + CONN\_ID)

| Column               | Data Type | Description                                                                                                                                                                                                                                                              |
|----------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hostname             | STRING    | Hostname data collected from                                                                                                                                                                                                                                             |
| agent_id             | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install.                                                                                                                                                                  |
| conn_id              | STRING    | Hash of "normalized" 5 tuple (ip1, port1, ip2, port2, L4 protocol). To represent A to B the same as B to A, the lower IP (as int) along with its port comes first in the hash preimage ordering. When ip1 == ip2, then the pair order is decided by lowest port instead. |
| first_seen           | TIMESTAMP | Earliest time seen                                                                                                                                                                                                                                                       |
| last_seen            | TIMESTAMP | Latest time seen                                                                                                                                                                                                                                                         |
| local_ip_addr        | STRING    | IP address on the host collecting data. This address is local to the host sensor. In dotted quad notation.                                                                                                                                                               |
| local_port           | INTEGER   | Port of the process on the collecting host                                                                                                                                                                                                                               |
| num_raw_rows         | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                                                                                                                   |
| os_family            | STRING    | OS Family: windows, linux, osx                                                                                                                                                                                                                                           |
| pid_hash             | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                                                                                                                                                                                                |
| process_name         | STRING    | Filename of executable                                                                                                                                                                                                                                                   |
| protocol             | STRING    | Layer 4 Protocol (TCP, UDP)                                                                                                                                                                                                                                              |
| remote_ip_addr       | STRING    | IP address the host collecting data is talking to. This address is remote to the host sensor. In dotted quad notation.                                                                                                                                                   |
| remote_port          | INTEGER   | Port of the remote host for this connection increment                                                                                                                                                                                                                    |
| sq_size              | DOUBLE    | Squared size of bytes in network packet. Ask Chris B!                                                                                                                                                                                                                    |
| sq_tcp_recv_size     | DOUBLE    | Square of the values in the interval. (Ask Chris B)                                                                                                                                                                                                                      |
| sq_tcp_send_size     | DOUBLE    | Square of the values in the interval. (Ask Chris B)                                                                                                                                                                                                                      |
| sq_udp_recv_size     | DOUBLE    | Square of the values in the interval. (Ask Chris B)                                                                                                                                                                                                                      |
| sq_udp_send_size     | DOUBLE    | Square of the values in the interval. (Ask Chris B)                                                                                                                                                                                                                      |
| tcp_accept_count     | DOUBLE    | Number of TCP ACCEPT events on this connection                                                                                                                                                                                                                           |
| tcp_connect_count    | DOUBLE    | Number of TCP CONNECT events on this connection                                                                                                                                                                                                                          |
| tcp_disconnect_count | DOUBLE    | Number of TCP DISCONNECT events on this connection                                                                                                                                                                                                                       |
| tcp_reconnect_count  | DOUBLE    | Number of TCP RECONNECT events on this connection                                                                                                                                                                                                                        |
| tcp_recv_count       | DOUBLE    | Number of TCP RECV events on this connection                                                                                                                                                                                                                             |
| tcp_recv_size        | DOUBLE    | Number of TCP RECV bytes received on this connection                                                                                                                                                                                                                     |
| tcp_retransmit_count | DOUBLE    | Number of TCP RETRANSMIT events on this connection                                                                                                                                                                                                                       |
| tcp_send_count       | DOUBLE    | Number of TCP SEND events on this connection                                                                                                                                                                                                                             |

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TABLE 20 – Continued from previous page

| Column            | Data Type | Description                                     |
|-------------------|-----------|-------------------------------------------------|
| tcp_send_size     | DOUBLE    | Number of TCP bytes sent                        |
| tcp_tcpcopy_count | DOUBLE    | Number of TCP TCPCOPY events on this connection |
| tcp_tcpcopy_size  | DOUBLE    | Number of TCP TCPCOPY bytes on this connection  |
| total_events      | DOUBLE    | Sum of events counts for this connection        |
| total_size        | DOUBLE    | Number of bytes observed for this connection    |
| udp_recv_count    | DOUBLE    | Number of UDP RECV events on this connection    |
| udp_recv_size     | DOUBLE    | Number of UDP bytes received                    |
| udp_send_count    | DOUBLE    | Number of UDP SEND events on this connection    |
| udp_send_size     | DOUBLE    | Number of UDP bytes sent on this connection     |

## 1.21. process\_net\_summary

Summarizes all network connections for a given process to the process level.

| Column               | Data Type | Description                                                                                             |
|----------------------|-----------|---------------------------------------------------------------------------------------------------------|
| Hostname             | STRING    | Hostname data collected from                                                                            |
| agent_id             | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| avg_bytes            | DOUBLE    | Average bytes in an event                                                                               |
| avg_packets          | DOUBLE    | Average packets per session done by the process                                                         |
| conn_id_count        | INT64     | Number of unique network connection IDs (5-tuples)                                                      |
| first_seen           | TIMESTAMP | Earliest time seen                                                                                      |
| last_seen            | TIMESTAMP | Latest time seen                                                                                        |
| max_bytes            | DOUBLE    | Maximum bytes in an event                                                                               |
| max_packets          | DOUBLE    | Max packets per session done by the process                                                             |
| min_bytes            | DOUBLE    | Minimum bytes in an event                                                                               |
| min_packets          | DOUBLE    | Min packets per session done by the process                                                             |
| net_recv_size        | DOUBLE    | Total bytes received by the process                                                                     |
| net_rs_total         | DOUBLE    | Total bytes sent/received by the process                                                                |
| net_send_size        | DOUBLE    | Total bytes sent by the process                                                                         |
| net_send_vs_recv     | DOUBLE    | Ratio of bytes sent vs received by the process                                                          |
| net_total_events     | DOUBLE    | Total events accross UDP/TCP                                                                            |
| net_total_size       | DOUBLE    | Total bytes accross all events                                                                          |
| num_raw_rows         | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.  |
| os_family            | STRING    | OS Family: windows, linux, osx                                                                          |
| pid_hash             | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                               |
| process_name         | STRING    | Filename of executable                                                                                  |
| sq_size              | DOUBLE    | Squared size of bytes in network packet. Ask Chris B!                                                   |
| tcp_accept_count     | DOUBLE    | Number of TCP Accept for this process                                                                   |
| tcp_connect_count    | DOUBLE    | Number of TCP Connects for this process                                                                 |
| tcp_disconnect_count | DOUBLE    | Number of TCP Disconnects for this process                                                              |
| tcp_reconnect_count  | DOUBLE    | Number of TCP Reconnects for this process                                                               |
| tcp_recv_count       | DOUBLE    | Number of TCP packets received by the process                                                           |
| tcp_recv_size        | DOUBLE    | Number of TCP bytes received by the process                                                             |
| tcp_retransmit_count | DOUBLE    | Number of TCP retransmits by the process                                                                |
| tcp_rs_total         | DOUBLE    | Total TCP bytes sent/received by the process                                                            |
| tcp_send_count       | DOUBLE    | Number of TCP packets sent by the process                                                               |
| tcp_send_size        | DOUBLE    | Number of TCP bytes sent by the process                                                                 |
| tcp_send_vs_recv     | DOUBLE    | Ratio of TCP bytes sent vs received by the process                                                      |
| tcp_tcpcopy_count    | DOUBLE    | Number of TCP copy events during the process                                                            |
| tcp_tcpcopy_size     | DOUBLE    | TCP copy bytes                                                                                          |

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TABLE 21 – Continued from previous page

| Column          | Data Type | Description                                        |
|-----------------|-----------|----------------------------------------------------|
| udp_rcv_count   | DOUBLE    | Number of UDP packets received by the process      |
| udp_rcv_size    | DOUBLE    | Number of UDP bytes received by the process        |
| udp_rs_total    | DOUBLE    | Total UDP bytes received by the process            |
| udp_send_count  | DOUBLE    | Number of UDP packets sent by the process          |
| udp_send_size   | DOUBLE    | Number of UDP bytes sent by the process            |
| udp_send_vs_rcv | DOUBLE    | Ratio of UDP bytes sent vs received by the process |

## 1.22. process\_path

For each process, has the path to its root (parent process).

| Column            | Data Type | Description                                                                                                                                     |
|-------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| agent_id          | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install.                                         |
| hostname          | STRING    | Hostname data collected from                                                                                                                    |
| level             | INTEGER   | Distance from the kernel process                                                                                                                |
| max_level         | INTEGER   | Ignore: The same value level. Artifact of the processing to generate the paths.                                                                 |
| os_pid            | INTEGER   | Operating system process id                                                                                                                     |
| parent_os_pid     | INTEGER   | OS Process ID of parent process                                                                                                                 |
| parent_pid_hash   | STRING    | PID_HASH of parent process                                                                                                                      |
| pid_hash          | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                                                                       |
| process_name      | STRING    | Filename of executable                                                                                                                          |
| process_path      | STRING    | Fully qualified path of the process executable                                                                                                  |
| ptree             | STRING    | Path to kernel using only PROCESS_NAMES. Order and format is: =process-¿parent process-¿... Ex: =winlogon.exe-¿smss.exe-¿smss.exe-¿ntoskrnl.exe |
| ptree_list        | STRING[]  | Path to kernel using PID_HASHes. Stored as an list. List is order from process to kernel process                                                |
| ptree_list_tuples | STRUCT    | Path to kernel using named tuples of PID_HASH and PROCESS_NAME (list of maps). List is order from process to kernel process                     |
| seq               | INTEGER   | Ignore: Artifact left over from process. Level+1.                                                                                               |

## 1.23. process\_registry

Registry activity events. These are aggregated to PROCESS + Registry Key/Value

| Column        | Data Type | Description                                                                                             |
|---------------|-----------|---------------------------------------------------------------------------------------------------------|
| activity_type | STRING    | Type of registry activity. Ex: CREATEKEY, READ, WRITE, DELETEVALUE, DELETEKEY                           |
| agent_id      | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| event_count   | DOUBLE    | Total ETW events                                                                                        |
| first_seen    | TIMESTAMP | Earliest time seen                                                                                      |
| hostname      | STRING    | Hostname data collected from                                                                            |
| last_seen     | TIMESTAMP | Latest time seen                                                                                        |
| max_event     | TIMESTAMP | Maximum value in the interval                                                                           |
| min_event     | TIMESTAMP | Minimum value in the interval                                                                           |

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TABLE 23 – Continued from previous page

| Column       | Data Type | Description                                                                                            |
|--------------|-----------|--------------------------------------------------------------------------------------------------------|
| num_raw_rows | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| pid_hash     | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                              |
| process_name | STRING    | Filename of executable                                                                                 |
| reg_data     | STRING    | Registry data, which is the most detailed part of the registry key. Strangely, it isn't data           |
| reg_path     | STRING    | Registry path                                                                                          |
| reg_value    | STRING    | Registry value read or written                                                                         |

#### 1.24. process\_registry\_summary

All registry activity events summarized to the PROCESS.

| Column               | Data Type | Description                                                                                             |
|----------------------|-----------|---------------------------------------------------------------------------------------------------------|
| agent_id             | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| createkeys           | DOUBLE    | Number of createkeys by process                                                                         |
| deletekeys           | DOUBLE    | Number of deletekeys by process                                                                         |
| deletevalues         | DOUBLE    | Number of deletevalues by process                                                                       |
| first_seen           | TIMESTAMP | Earliest time seen                                                                                      |
| hostname             | STRING    | Hostname data collected from                                                                            |
| last_seen            | TIMESTAMP | Latest time seen                                                                                        |
| pid_hash             | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                               |
| process_name         | STRING    | Filename of executable                                                                                  |
| reads                | DOUBLE    | Number of reads by process                                                                              |
| total_activity_types | DOUBLE    | Total number of ETW events for all registry activity for a process                                      |
| writes               | DOUBLE    | Number of writes by process                                                                             |

#### 1.25. process\_summary

Provides a unified summary of all process-related activities that are directly collected by Wintap. This includes:

- Process metadata (e.g., process\_name, hostname, pid\_hash).
- Registry activity (e.g., reads, writes, key creation/deletion).
- File activity (e.g., file reads, writes, and other events).
- Network activity (e.g., connections, data sent/received).
- Image loads (e.g., DLLs loaded by the process).
- Host-level metadata (e.g., OS, architecture).

| Column         | Data Type | Description                                                                                             |
|----------------|-----------|---------------------------------------------------------------------------------------------------------|
| Close_Events   | DOUBLE    | Number of file close events                                                                             |
| Create_Events  | DOUBLE    | Number of file create events                                                                            |
| Delete_Events  | DOUBLE    | Number of file delete events                                                                            |
| Read_Bytes     | DOUBLE    | File bytes read                                                                                         |
| Read_Events    | DOUBLE    | Number of file read events                                                                              |
| Rename_Events  | DOUBLE    | Number of file rename events                                                                            |
| SetInfo_Events | DOUBLE    |                                                                                                         |
| Write_Bytes    | DOUBLE    | File bytes written                                                                                      |
| Write_Events   | DOUBLE    | Number of file write events                                                                             |
| agent_id       | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install. |
| arch           | STRING    | Host architecture                                                                                       |

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TABLE 25 – Continued from previous page

| Column             | Data Type | Description                                                                                                                                                                   |
|--------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| args               | STRING    | Process arguments. Note: as a simple string, no parsing done                                                                                                                  |
| avg_bytes          | DOUBLE    | Average bytes over the interval                                                                                                                                               |
| avg_packets        | DOUBLE    | Average packets over the interval                                                                                                                                             |
| commit_charge      | INT64     | Commit charge is the total amount of virtual memory guaranteed for all processes to fit in physical memory and the page file.                                                 |
| commit_peak        | INT64     | Total memory committed                                                                                                                                                        |
| conn_id_count      | INT64     | Number of unique network connection IDs (5-tuples)                                                                                                                            |
| cpu_cycle_count    | INT64     | Always zero (note: more than zero cycles were generally used in a process)                                                                                                    |
| cpu_utilization    | INTEGER   | Ignore: unable to collect. This *should* be a value indicating the percentage of cpu utilized by the process over its life, but turns out be harder to collect than expected. |
| dll_first_seen     | TIMESTAMP | Earliest time seen                                                                                                                                                            |
| dll_last_seen      | TIMESTAMP | Latest time seen                                                                                                                                                              |
| dll_num_uniq_files | INT64     | Number of unique DDLs by name                                                                                                                                                 |
| dlls               | STRING[]  | List of DDL names as an array                                                                                                                                                 |
| duration_seconds   | DOUBLE    | Elapsed process execution in seconds                                                                                                                                          |
| exit_code          | INT64     | Process exit code                                                                                                                                                             |
| file_first_seen    | TIMESTAMP | Earliest time seen                                                                                                                                                            |
| file_id            | STRING    | Unique ID for a file. Hash of hostname+fully qualified filename                                                                                                               |
| file_last_seen     | TIMESTAMP | Latest time seen                                                                                                                                                              |
| file_md5           | STRING    | MD5 hash of the binary process/dll                                                                                                                                            |
| file_num_raw_rows  | DOUBLE    | Number of rows of file activity from the raw data.                                                                                                                            |
| file_sha2          | STRING    | SHA2 hash of the file contents                                                                                                                                                |
| filename           | STRING    | Fully qualified filename                                                                                                                                                      |
| first_seen         | TIMESTAMP | Earliest time seen                                                                                                                                                            |
| hard_fault_count   | INTEGER   | Count of hard page faults                                                                                                                                                     |
| hostname           | STRING    | Hostname data collected from                                                                                                                                                  |
| last_seen          | TIMESTAMP | Latest time seen                                                                                                                                                              |
| max_bytes          | DOUBLE    | Max bytes in the interval                                                                                                                                                     |
| max_packets        | DOUBLE    | Max packets in the interval                                                                                                                                                   |
| min_bytes          | DOUBLE    | Min bytes in the interval                                                                                                                                                     |
| min_packets        | DOUBLE    | Max packets in the interval                                                                                                                                                   |
| net_first_seen     | TIMESTAMP | Earliest time seen                                                                                                                                                            |
| net_last_seen      | TIMESTAMP | Latest time seen                                                                                                                                                              |
| net_num_raw_rows   | DOUBLE    | Number of rows in the raw data                                                                                                                                                |
| net_recv_size      | DOUBLE    | Total bytes received in the interval                                                                                                                                          |
| net_rs_total       | DOUBLE    | Total bytes sent/received in the interval                                                                                                                                     |
| net_send_size      | DOUBLE    | Total bytes sent in the interval                                                                                                                                              |
| net_send_vs_recv   | DOUBLE    | Ratio of bytes sent/received in the interval                                                                                                                                  |
| net_total_events   | DOUBLE    | Total network events in the interval                                                                                                                                          |
| net_total_size     | DOUBLE    | Total bytes (send/receive) in the interval                                                                                                                                    |
| num_agent_id       | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                        |
| num_args           | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                        |
| num_file_md5       | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                        |
| num_file_sha2      | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                        |
| num_null_filename  | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                        |
| num_parent_os_pid  | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value.                                                                        |

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TABLE 25 – Continued from previous page

| Column                  | Data Type | Description                                                                                            |
|-------------------------|-----------|--------------------------------------------------------------------------------------------------------|
| num_parent_pid_hash     | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_name        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_path        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_start       | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_stop        | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_uniq_file_hash      | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_user_name           | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| os                      | STRING    | Operating System                                                                                       |
| os_family               | STRING    | OS Family: windows, linux, osx                                                                         |
| os_pid                  | INTEGER   | Operating system process id                                                                            |
| os_version              | STRING    | Operating System Version                                                                               |
| parent_os_pid           | INTEGER   | OS Process ID of parent process                                                                        |
| parent_pid_hash         | STRING    | PID_HASH of parent process                                                                             |
| pid_hash                | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                              |
| process_name            | STRING    | Filename of executable                                                                                 |
| process_path            | STRING    | Fully qualified path of the process executable                                                         |
| process_started         | TIMESTAMP | Process start timestamp                                                                                |
| process_started_seconds | DOUBLE    | Process start time in unix epoch seconds                                                               |
| process_stop_seconds    | DOUBLE    | Process stop in unix epoch seconds                                                                     |
| process_term            | TIMESTAMP | Process termination timestamp                                                                          |
| read_operation_count    | INT64     | Count of number of read operations                                                                     |
| read_transfer_kilobytes | INT64     | Number of K bytes read                                                                                 |
| reg_createkeys          | DOUBLE    | Registry keys created                                                                                  |
| reg_deletekeys          | DOUBLE    | Registry keys deleted                                                                                  |
| reg_deletevalues        | DOUBLE    | Registry values deleted                                                                                |
| reg_first_seen          | TIMESTAMP | Earliest time seen                                                                                     |
| reg_last_seen           | TIMESTAMP | Latest time seen                                                                                       |
| reg_reads               | DOUBLE    | Registry read events                                                                                   |
| reg_totals              | DOUBLE    | Total registry events                                                                                  |
| reg_writes              | DOUBLE    | Registry write events                                                                                  |
| sq_size                 | DOUBLE    | Squared size of bytes in network packet. Ask Chris B!                                                  |
| tcp_accept_count        | DOUBLE    | TCP Accepts                                                                                            |
| tcp_connect_count       | DOUBLE    | TCP connects                                                                                           |
| tcp_disconnect_count    | DOUBLE    | TCP disconnects                                                                                        |
| tcp_reconnect_count     | DOUBLE    | TCP reconnects                                                                                         |
| tcp_recv_count          | DOUBLE    | TCP packets received                                                                                   |
| tcp_recv_size           | DOUBLE    | TCP bytes received                                                                                     |
| tcp_retransmit_count    | DOUBLE    | TCP packets retransmitted                                                                              |
| tcp_rs_total            | DOUBLE    | TCP bytes total (send/received)                                                                        |
| tcp_send_count          | DOUBLE    | TCP bytes sent                                                                                         |
| tcp_send_size           | DOUBLE    | TCP packets sent                                                                                       |
| tcp_send_vs_recv        | DOUBLE    | Ratio of TCP bytes sent/received                                                                       |
| tcp_tcpcopy_count       | DOUBLE    | TCP copy events. We think this is an event that is internal to the network stack on the host.          |
| tcp_tcpcopy_size        | DOUBLE    | TCP copy bytes                                                                                         |

Continued on next page

TABLE 25 – Continued from previous page

| Column                   | Data Type | Description                                                                                                                                                                                                                                                                                                |
|--------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| token_elevation_type     | INTEGER   | Indicates UAC privilege status: 1 (Default) = no linked token, UAC disabled or non-admin account; 2 (Full) = elevated administrator with full rights after UAC approval; 3 (Limited) = filtered admin token before elevation with available full token; 4 = undocumented value, treat as unknown/reserved. |
| udp_rcv_count            | DOUBLE    | UDP bytes received                                                                                                                                                                                                                                                                                         |
| udp_rcv_size             | DOUBLE    | UDP events received                                                                                                                                                                                                                                                                                        |
| udp_rs_total             | DOUBLE    | Total UDP bytes sent/received                                                                                                                                                                                                                                                                              |
| udp_send_count           | DOUBLE    | UDP packets sent                                                                                                                                                                                                                                                                                           |
| udp_send_size            | DOUBLE    | UDP bytes sent                                                                                                                                                                                                                                                                                             |
| udp_send_vs_rcv          | DOUBLE    | Ratio of UDP bytes sent/recieved                                                                                                                                                                                                                                                                           |
| user_name                | STRING    | Username. Note that on windows there are many usernames that represent system or other background activity                                                                                                                                                                                                 |
| write_operation_count    | INT64     | Count of number of write operations                                                                                                                                                                                                                                                                        |
| write_transfer_kilobytes | INT64     | Number of K bytes written                                                                                                                                                                                                                                                                                  |

## 1.26. process\_uber\_summary

The process\_uber\_summary view is an enhanced version of the process\_summary view, incorporating additional threat intelligence and labeling data. It combines process activity summaries with external threat indicators such as SIGMA rules, MITRE ATT&CK techniques, LOLBAS (Living Off the Land Binaries and Scripts), and NetworkX graph labels. This comprehensive view provides a detailed and enriched dataset for advanced threat analysis and detection.

| Column                  | Data Type | Description                                                                                                                                                                   |
|-------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Close_Events            | DOUBLE    | Number of file close events                                                                                                                                                   |
| Create_Events           | DOUBLE    | Number of file create events                                                                                                                                                  |
| Delete_Events           | DOUBLE    | Number of file delete events                                                                                                                                                  |
| Read_Bytes              | DOUBLE    | File bytes read                                                                                                                                                               |
| Read_Events             | DOUBLE    | Number of file read events                                                                                                                                                    |
| Rename_Events           | DOUBLE    | Number of file rename events                                                                                                                                                  |
| SetInfo_Events          | DOUBLE    |                                                                                                                                                                               |
| Write_Bytes             | DOUBLE    | File bytes written                                                                                                                                                            |
| Write_Events            | DOUBLE    | Number of file write events                                                                                                                                                   |
| agent_id                | STRING    | Agent ID is a unique id for a Wintap install on a host. It is created on initial startup after install.                                                                       |
| arch                    | STRING    | Host architecture                                                                                                                                                             |
| args                    | STRING    | Process arguments. Note: as a simple string, no parsing done                                                                                                                  |
| avg_bytes               | DOUBLE    | Average bytes over the interval                                                                                                                                               |
| avg_packets             | DOUBLE    | Average packets over the interval                                                                                                                                             |
| commit_charge           | INT64     | Commit charge is the total amount of virtual memory guaranteed for all processes to fit in physical memory and the page file.                                                 |
| commit_peak             | INT64     | Total memory committed                                                                                                                                                        |
| conn_id_count           | INT64     | Number of unique network connection IDs (5-tuples)                                                                                                                            |
| cpu_cycle_count         | INT64     | Always zero (note: more than zero cycles were generally used in a process)                                                                                                    |
| cpu_utilization         | INTEGER   | Ignore: unable to collect. This *should* be a value indicating the percentage of cpu utilized by the process over its life, but turns out be harder to collect than expected. |
| critical_num_sigma_hits | DOUBLE    | Number of unique Sigma rules hit                                                                                                                                              |
| dll_first_seen          | TIMESTAMP | Earliest time seen                                                                                                                                                            |
| dll_last_seen           | TIMESTAMP | Latest time seen                                                                                                                                                              |
| dll_num_uniq_files      | INT64     | Number of unique DDLs by name                                                                                                                                                 |

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TABLE 26 – Continued from previous page

| Column                     | Data Type  | Description                                                                                            |
|----------------------------|------------|--------------------------------------------------------------------------------------------------------|
| dlls                       | STRING[]   | List of DDL names as an array                                                                          |
| duration_seconds           | DOUBLE     | Elapsed process execution in seconds                                                                   |
| exit_code                  | INT64      | Process exit code                                                                                      |
| file_first_seen            | TIMESTAMP  | Earliest time seen                                                                                     |
| file_id                    | STRING     | Unique ID for a file. Hash of hostname+fully qualified filename                                        |
| file_last_seen             | TIMESTAMP  | Latest time seen                                                                                       |
| file_md5                   | STRING     | MD5 hash of the binary process/dll                                                                     |
| file_num_raw_rows          | DOUBLE     | Number of rows of file activity from the raw data.                                                     |
| file_sha2                  | STRING     | SHA2 hash of the file contents                                                                         |
| filename                   | STRING     | Fully qualified filename                                                                               |
| first_seen                 | TIMESTAMP  | Earliest time seen                                                                                     |
| hard_fault_count           | INTEGER    | Count of hard page faults                                                                              |
| high_num_sigma_hits        | INT64      | Number of unique Sigma rules hit                                                                       |
| high_num_sigma_rows        | INT64      | Number of times Sigma rules hit for this entity                                                        |
| hostname                   | STRING     | Hostname data collected from                                                                           |
| label_num_hits             | INT64      | Number of unique labels hit                                                                            |
| label_num_sources          | INT64      | Number of unique sources of labels                                                                     |
| label_num_uniq_annotations | INT64      | Number of unique annotations                                                                           |
| label_source               | STRING     | For now, only a single source: "networkx"                                                              |
| last_seen                  | TIMESTAMP  | Latest time seen                                                                                       |
| lolbas_cats                | STRING[]   | List categories from hits                                                                              |
| lolbas_mitre               | STRING[]   | List of Mitre codes from hits                                                                          |
| lolbas_num_rows            | INT64      | Number of hits                                                                                         |
| lolbas_privs               | STRING[]   | List of privileges from hits                                                                           |
| low_num_sigma_hits         | INT64      | Number of unique Sigma rules hit                                                                       |
| low_num_sigma_rows         | INT64      | Number of times Sigma rules hit for this entity                                                        |
| max_bytes                  | DOUBLE     | Max bytes in the interval                                                                              |
| max_packets                | DOUBLE     | Max packets in the interval                                                                            |
| medium_num_sigma_hits      | INT64      | Number of unique Sigma rules hit                                                                       |
| medium_num_sigma_rows      | INT64      | Number of times Sigma rules hit for this entity                                                        |
| min_bytes                  | DOUBLE     | Min bytes in the interval                                                                              |
| min_packets                | DOUBLE     | Max packets in the interval                                                                            |
| mitre_analytic_ids         | STRING[]   | List of analytic IDs from hits                                                                         |
| mitre_analytic_types       | STRING[][] | List of analytic types from hits                                                                       |
| mitre_information_domains  | STRING[]   | List domains from hits. Ex: Analytic, Host, Network                                                    |
| mitre_num_rows             | INT64      | Number of hits                                                                                         |
| mitre_subtypes             | STRING[][] | List of subtypes from hits. Ex: Map building, Anomaly, Host-flow, Process.                             |
| net_first_seen             | TIMESTAMP  | Earliest time seen                                                                                     |
| net_last_seen              | TIMESTAMP  | Latest time seen                                                                                       |
| net_num_raw_rows           | DOUBLE     | Number of rows in the raw data                                                                         |
| net_recv_size              | DOUBLE     | Total bytes received in the interval                                                                   |
| net_rs_total               | DOUBLE     | Total bytes sent/received in the interval                                                              |
| net_send_size              | DOUBLE     | Total bytes sent in the interval                                                                       |
| net_send_vs_recv           | DOUBLE     | Ratio of bytes sent/received in the interval                                                           |
| net_total_events           | DOUBLE     | Total network events in the interval                                                                   |
| net_total_size             | DOUBLE     | Total bytes (send/receive) in the interval                                                             |
| num_agent_id               | INT64      | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_args                   | INT64      | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_file_md5               | INT64      | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_file_sha2              | INT64      | Number of unique values in the related column. Typically, the related column should only have 1 value. |

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TABLE 26 – Continued from previous page

| Column                  | Data Type | Description                                                                                            |
|-------------------------|-----------|--------------------------------------------------------------------------------------------------------|
| num_null_filename       | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_parent_os_pid       | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_parent_pid_hash     | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_name        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_path        | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_start       | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_process_stop        | DOUBLE    | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_uniq_file_hash      | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| num_user_name           | INT64     | Number of unique values in the related column. Typically, the related column should only have 1 value. |
| os                      | STRING    | Operating System                                                                                       |
| os_family               | STRING    | OS Family: windows, linux, osx                                                                         |
| os_pid                  | INTEGER   | Operating system process id                                                                            |
| os_version              | STRING    | Operating System Version                                                                               |
| parent_os_pid           | INTEGER   | OS Process ID of parent process                                                                        |
| parent_pid_hash         | STRING    | PID_HASH of parent process                                                                             |
| pid_hash                | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time)                              |
| process_name            | STRING    | Filename of executable                                                                                 |
| process_path            | STRING    | Fully qualified path of the process executable                                                         |
| process_started         | TIMESTAMP | Process start timestamp                                                                                |
| process_started_seconds | DOUBLE    | Process start time in unix epoch seconds                                                               |
| process_stop_seconds    | DOUBLE    | Process stop in unix epoch seconds                                                                     |
| process_term            | TIMESTAMP | Process termination timestamp                                                                          |
| read_operation_count    | INT64     | Count of number of read operations                                                                     |
| read_transfer_kilobytes | INT64     | Number of K bytes read                                                                                 |
| reg_createkeys          | DOUBLE    | Registry keys created                                                                                  |
| reg_deletekeys          | DOUBLE    | Registry keys deleted                                                                                  |
| reg_deletevalues        | DOUBLE    | Registry values deleted                                                                                |
| reg_first_seen          | TIMESTAMP | Earliest time seen                                                                                     |
| reg_last_seen           | TIMESTAMP | Latest time seen                                                                                       |
| reg_reads               | DOUBLE    | Registry read events                                                                                   |
| reg_totals              | DOUBLE    | Total registry events                                                                                  |
| reg_writes              | DOUBLE    | Registry write events                                                                                  |
| sq_size                 | DOUBLE    | Squared size of bytes in network packet. Ask Chris B!                                                  |
| tcp_accept_count        | DOUBLE    | TCP Accepts                                                                                            |
| tcp_connect_count       | DOUBLE    | TCP connects                                                                                           |
| tcp_disconnect_count    | DOUBLE    | TCP disconnects                                                                                        |
| tcp_reconnect_count     | DOUBLE    | TCP reconnects                                                                                         |
| tcp_rcv_count           | DOUBLE    | TCP packets received                                                                                   |
| tcp_rcv_size            | DOUBLE    | TCP bytes received                                                                                     |
| tcp_retransmit_count    | DOUBLE    | TCP packets retransmitted                                                                              |
| tcp_rs_total            | DOUBLE    | TCP bytes total (send/received)                                                                        |
| tcp_send_count          | DOUBLE    | TCP bytes sent                                                                                         |
| tcp_send_size           | DOUBLE    | TCP packets sent                                                                                       |
| tcp_send_vs_rcv         | DOUBLE    | Ratio of TCP bytes sent/received                                                                       |

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TABLE 26 – Continued from previous page

| Column                   | Data Type | Description                                                                                                                                                                                                                                                                                                |
|--------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tcp_tcpcopy_count        | DOUBLE    | TCP copy events. We think this is an event that is internal to the network stack on the host.                                                                                                                                                                                                              |
| tcp_tcpcopy_size         | DOUBLE    | TCP copy bytes                                                                                                                                                                                                                                                                                             |
| token_elevation_type     | INTEGER   | Indicates UAC privilege status: 1 (Default) = no linked token, UAC disabled or non-admin account; 2 (Full) = elevated administrator with full rights after UAC approval; 3 (Limited) = filtered admin token before elevation with available full token; 4 = undocumented value, treat as unknown/reserved. |
| total_sigma_hits         | DOUBLE    | Total Sigma hits over all criticality levels                                                                                                                                                                                                                                                               |
| udp_rcv_count            | DOUBLE    | UDP bytes received                                                                                                                                                                                                                                                                                         |
| udp_rcv_size             | DOUBLE    | UDP events received                                                                                                                                                                                                                                                                                        |
| udp_rs_total             | DOUBLE    | Total UDP bytes sent/received                                                                                                                                                                                                                                                                              |
| udp_send_count           | DOUBLE    | UDP packets sent                                                                                                                                                                                                                                                                                           |
| udp_send_size            | DOUBLE    | UDP bytes sent                                                                                                                                                                                                                                                                                             |
| udp_send_vs_rcv          | DOUBLE    | Ratio of UDP bytes sent/received                                                                                                                                                                                                                                                                           |
| user_name                | STRING    | Username. Note that on windows there are many usernames that represent system or other background activity                                                                                                                                                                                                 |
| write_operation_count    | INT64     | Count of number of write operations                                                                                                                                                                                                                                                                        |
| write_transfer_kilobytes | INT64     | Number of K bytes written                                                                                                                                                                                                                                                                                  |

## 1.27. sigma\_labels

The sigma\_labels table is a key component in the threat detection pipeline, as it links process or entity activity to specific SIGMA rules. SIGMA is a standardized rule format for describing log-based detection patterns, often used for identifying suspicious or malicious behavior in systems. The sigma\_labels table provides a mapping between entities (like processes) and the SIGMA rules that matched their activity, along with metadata about the severity and type of the detection.

| Column      | Data Type | Description                                                                                                                                                 |
|-------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| analytic_id | STRING    | Specific SIGMA rule hit. Value is a GUID                                                                                                                    |
| entity      | STRING    | Wintap entity ID (hash). The type of entity (PID_HASH, CONN_ID, etc) is defined in the ENTITY_TYPE column. Note: Currently, only PID_HASHes are identified. |
| entity_type | STRING    | Wintap entity type. Determines what kind of hash is in the ENTITY column. Ex: PID_HASH, CONN_ID, FILE_ID                                                    |
| time        | TIMESTAMP | Timestamp of activity                                                                                                                                       |

## 1.28. sigma\_labels\_summary

Aggregates data from sigma\_labels by PID\_HASH and pivots the severity levels into separate columns (e.g., critical\_num\_sigma\_hits, high\_num\_sigma\_hits).

| Column                  | Data Type | Description                                                               |
|-------------------------|-----------|---------------------------------------------------------------------------|
| critical_num_sigma_hits | DOUBLE    | Number of unique Sigma rules hit                                          |
| high_num_sigma_hits     | INT64     | Number of unique Sigma rules hit                                          |
| high_num_sigma_rows     | INT64     | Number of times Sigma rules hit for this entity                           |
| low_num_sigma_hits      | INT64     | Number of unique Sigma rules hit                                          |
| low_num_sigma_rows      | INT64     | Number of times Sigma rules hit for this entity                           |
| medium_num_sigma_hits   | INT64     | Number of unique Sigma rules hit                                          |
| medium_num_sigma_rows   | INT64     | Number of times Sigma rules hit for this entity                           |
| pid_hash                | STRING    | Globally unique process hash (hash from hostname, os pid, and start_time) |